

Aniruddha Chakraborty

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EDUCATION

Ph.D., Astronomy & Astrophysics — Tata Institute of Fundamental Research, [Year]–Present

Advisor: Prof. Suvodip Mukherjee — [thesis title, if applicable]

RESEARCH FOCUS

Lensing of gravitational waves — detection methodology and application prospects. Development of model-independent techniques to identify lensed gravitational-wave signals in both the geometric-optics and wave-optics regimes.

PUBLICATIONS

A. Chakraborty and S. Mukherjee. "GLANCE: Gravitational Lensing Authenticator using Non-modelled Cross-correlation Exploration of Gravitational Wave Signals." *MNRAS*, 532(4), 4842–4863, 2024. arXiv:2403.03982

A. Chakraborty and S. Mukherjee. " μ -GLANCE: A Novel Technique to Detect Chromatically and Achromatically Lensed Gravitational-wave Signals." *ApJ*, 984, 107, 2025. arXiv:2410.06995

SKILLS

Python, gravitational-wave data analysis (PyCBC, bilby), signal cross-correlation techniques, gravitational lensing modeling, p5.js